

PHYSICS

Used in glowing tubelight

Watt: SI unit of power

electrical power = Ratio of electrical energy 2 times

Electrical Energy
Time

Electrical Energy = Watt x hour

Electrical Energy = $\frac{1}{1000}$ kilowatt hr

= $\frac{1}{1000}$ KWH = 1 Unit

The mercury of 40W is working for 6 hrs

Electrical energy consumed during

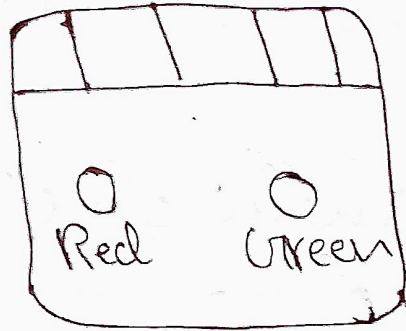
6 hrs = Power x time

= 40×6 Wh

= 240 Wh

= $\frac{240}{1000}$ KWH = 0.240 KWH

Domestic circuit.



1 kWh = 1 Unit
electrical energy

Electrical energy consumed during 6 hrs for 4 tubelight = 0.960 kWh

Electrical energy consumed during 6 hrs for 30 days.

$$\begin{aligned}
 &= (0.96 \times 30) \text{ kWh} \\
 &= 28.8 \text{ kWh} \\
 &= 28.8 \text{ Unit}
 \end{aligned}$$

The cost of 1 Unit = ₹5.80

$$\begin{array}{r}
 28.8 \\
 \times 5.8 \\
 \hline
 2304 \\
 1440 \\
 \hline
 \text{₹ } 167.04
 \end{array}$$

The cost of 28.8 Unit for 500 rooms = 500×167.04

$$= ₹ 83520$$

GEOGRAPHY

Climate of India

Tropical Monsoon Climate →

India lies between $8^{\circ}4'N$ to $37^{\circ}6'N$ latitudes with tropic of cancer passing through its middle giving India its tropical climate.

It experiences seasonal change in wind direction and therefore is named to have a monsoon climate which means season thus India has a tropical monsoon climate.

- 'India is a country of great climatic diversity' Justify.

MATHS

18/11/25

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Inverse Variation

- Two quantities a & b are said to vary inversely if the product ab remains constant

Example -

Suppose a car covers a dist. at a uniform speed. Then we can say for certain that more is the speed of the car, less is the time taken to cover this dist.

Thus, speed varies inversely as the time taken to cover a certain dist.

Rules for inverse variation:

If a & b vary inversely then the ratio of any two values of a is equal to the inverse ratio of the corresponding values of b .

Q) If 36 men can finish a piece of work in 25 days, how many days will 15 men take to do it?

36 men can finish a piece of work in 25 days, ~~how many days will 15 men~~

1 man can finish it in (25×36) days

$\therefore$ 15 men can finish it in

$$\frac{5 \cdot 12}{(25 \times 36)} \text{ days} = 60 \text{ days}$$

$\frac{45}{3}$

Hence the required no. of days = 60

Q) A contractor has a work force of 420 men who can finish a certain piece of work in 9 months. How many extra men must be employed to complete the job in 7 months?

To finish work in 9 months,
men req. = 420

To finish work in 1 month = (420×9)

To finish work in 7 months, men,
req. = $\frac{60}{7} \times (420 \times 9) = 540$

Hence the no. of extra men
req. is $(540 - 420) = 120$

★ Q) A garrison of 500 men had
provisions for 24 days. However,
a reinforcement of 300 men
arrived. For how many
days will the food last now?

New strength = $(500 + 300) = 800$ men.
For 500 men there is food for 24 days.
For 1 man there is food for $= \frac{(500 \times 24)}{500}$ days

For 800 men there is food for =
 $\frac{(500 \times 24)}{800}$ days = 15 days

Hence the food will last for 15 days for

EX-9B

- 1) If 48 men can dig a trench in 14 days, how long will 28 men take to dig the same trench?

48 men can dig a trench in 14 days.

1 man can dig the trench in (48×14) days

$\therefore$ 28 men can dig the trench in $\frac{48 \times 14}{28}$ days = 24 days

Hence 28 men can dig the trench in 24 days

- 2) 16 men can reap a field in 30 days. How many men must be engaged to reap the same field in 24 days?

men req. to reap the field in
30 days = 16 men

men req. to reap the field in
1 day = (16×30) days.

$\therefore$ men req. to reap the field in
24 days = $(16 \times 30) \div 24 = 20$ men

$$\begin{array}{r} 24 \\ \times 16 \\ \hline 144 \\ 240 \\ \hline 384 \end{array}$$

AI

Q Differentiate active & passive footprint

Active footprint

- Information shared by user

- Uses personal information shared by themselves.

Passive footprint

- Information traced by internet activities.

- Uses personal information stolen from internet activities

4) Write short notes on:

a) Phishing

Phishing is a type of an unethical activity on Internet in which some miscreants try to trap people with interesting messages and offers which might look attractive but are actually meant for cheating or hacking purposes.

where the user gets cheated and his personal or sensitive data such as bank account details, passwords etc. are hacked and misused.

b) Right to Privacy

The Right to Privacy includes the concept of protecting an individual's personal information intact from public or others.

c) Cyber Crime

Internet provides a huge and diversified platform to work.

Some people misuse it and perform illegal and inappropriate actions and get involved in piracy, hacking etc.

d) IPR

'IPR' is the legal right that covers or ensures the privileges of an individual who has made, created something by his own that no one can his work by anyone's name.

5) ATQ

c) Write few guidelines to be followed while using Internet.

- Use strong password that is combination of lowercase, uppercase, digit and special symbols.

- Keep your system updated.

- Install Firewall to prevent from any outsider entry.
- Don't click on links shared by unknown people.
- Do not accept friend request from unknown.

e) How internet is affecting inherent abilities of a person? Elaborate with an example:

- People are lagging with calculation abilities due to overuse of calculators.
- People can't able to memorize birth dates, special days of their life and using calendars to remember the dates.
- People are using google maps to locate any route due to which their ability to memorize routes are decreasing.

- Playing Online games for a longer period of time leads to several health issues.